

6. The main stages in the manufacture of sulfuric acid are given below.



Stage 3 production of sulfuric acid from sulfur trioxide

- (a) State the meaning of the symbol $\rightleftharpoons$ used in the **stage 2** equation. [1]

- (b) The temperature used in **stage 2** is 450 °C. This results in a good yield but a low rate.

Name the **compound** used as a catalyst to increase the rate. [1]

- (c) Dissolving sulfur trioxide in water to form sulfuric acid is too exothermic to be carried out safely in one step.

Give the **two** steps carried out in **stage 3** to convert sulfur trioxide safely into sulfuric acid. [2]

Step 1

Step 2



Examiner
only

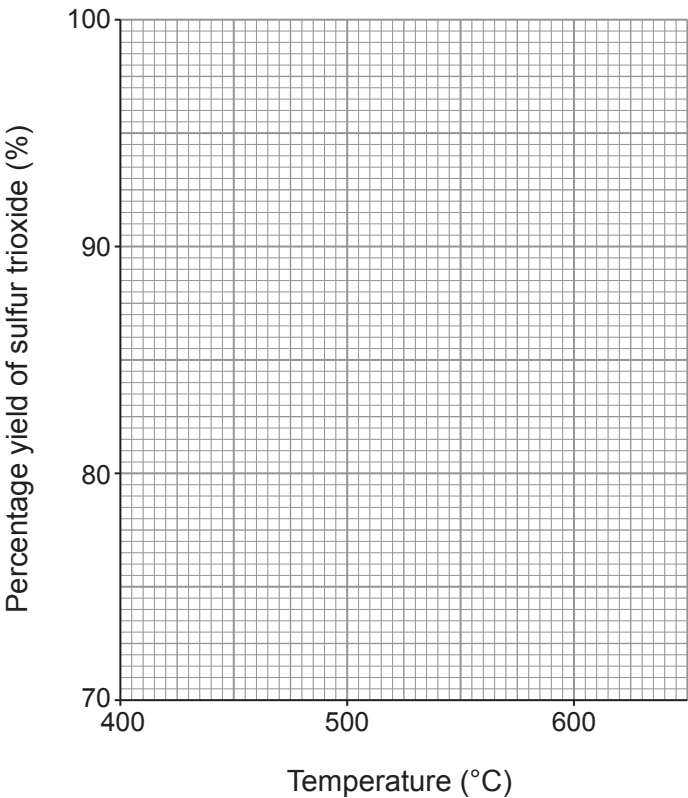
- (d) The catalyst used in **stage 2** will **not** work below 400 °C and **breaks down** above 620 °C.

The table shows the yield of sulfur trioxide at different temperatures.

Temperature (°C)	400	450	500	550	600
Percentage yield of sulfur trioxide (%)	99.0	96.5	92.5	86.0	76.5

Plot this data on the grid and draw a suitable line.

[3]



- (e) State what happens to the percentage yield of sulfur trioxide when the temperature is increased.

[1]

.....

- (f) Use your graph to give the temperature range that produces a yield of 90–99% of sulfur trioxide.

[1]

..... to °C

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